FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014: 2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

ALL INDIA INTEGRATED TEST SERIES

FIITJEE

JEE(Main), 2015

ANSWERS, HINTS & SOLUTIONS

HALF COURSE TEST–VI
(Main)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	**B**	**D**	**C**
2.	**A**	**B**	**B**
3.	**A**	**B**	**C**
4.	**D**	**B**	**A**
5.	**B**	**C**	**C**
6.	**A**	**C**	**D**
7.	**B**	**C**	**C**
8.	**D**	**D**	**C**
9.	**D**	**B**	**B**
10.	**D**	**D**	**B**
11.	**B**	**C**	**D**
12.	**C**	**A**	**C**
13.	**D**	**C**	**A**
14.	**D**	**C**	**B**
15.	**A**	**B**	**D**
16.	**B**	**C**	**B**
17.	**C**	**B**	**C**
18.	**C**	**B**	**B**
19.	**C**	**C**	**A**
20.	**C**	**D**	**A**
21.	**B**	**C**	**D**
22.	**A**	**C**	**C**
23.	**C**	**A**	**A**
24.	**B**	**C**	**C**
25.	**C**	**C**	**A**
26.	**D**	**D**	**B**
27.	**D**	**C**	**A**
28.	**A**	**B**	**D**
29.	**D**	**B**	**D**
30.	**B**	**C**	**B**

Physics

PART – I

SECTION – A

1. In sliding maximum friction will act where as in rolling static friction will act.

2. Conservation of angular momentum

$$mv\ell = \left(\frac{m\ell^2}{3} + m\ell^2\right)\omega \qquad (1)$$

conservation of energy

$$-mg\frac{\ell}{2} - mg\ell + \frac{1}{2}\left(\frac{m\ell^2}{3} + m\ell^2\right)\omega^2 = mg\frac{\ell}{2} + mg\ell \qquad (2)$$

$\Rightarrow \quad v = \sqrt{8g\ell}$

3. Apply conservation of angular momentum.

4. Cheek the distribution of mass about the three axis.

5. $I = \frac{m\ell^2}{3}$, $I' = \left[\frac{1}{12}\left(\frac{m}{2}\right)\left(\frac{\ell}{2}\right)^2 + \frac{m}{2}\left(\frac{5\ell^2}{16}\right)\right] + \frac{1}{3}\left(\frac{m}{2}\right)\left(\frac{\ell}{2}\right)^2 = \frac{5}{8}I$

6. acceleration a = g/2

velocity after 1 sec $v = \frac{g}{2}\times 1 = \frac{g}{2}$ m/s

the string will given become taut when $\frac{g}{2}t = \frac{1}{2}gt^2$

$\Rightarrow \quad t = 1$ sec.

7. $V_{m/M} = V_M \qquad (1)$

$V_m^2 = V_{m/M}^2 + V_M^2 + 2V_{m/M}V_M\cos 120°$

$\Rightarrow \quad V_m / V_M = 1$

8. $a = \frac{\text{spring force} - mg}{m_1} = \frac{m_2 g - m_1 g}{m_1}$

9. From the graph it can be concluded that both S_1 & S_2 are accelerated frame or either S_1 or S_2 is accelerated.

10. The tension in the string is zero and block is falling freely.

11. Conservation of linear momentum

$mu = (M + m)V \Rightarrow V = \frac{mu}{M+m}$

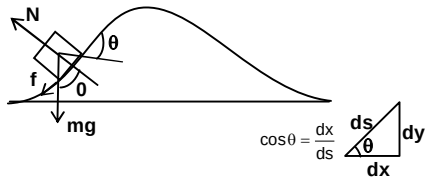
12. Using conservation of energy and $F = \frac{mu}{M+m}$

14. Angular velocity and angular acceleration are parallel or antiparallel in a circular motion.

16. Displacement time curve will be a parabola.

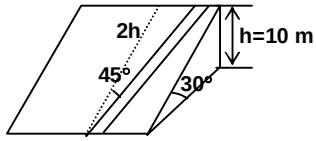
17. Work done by friction $= \int \vec{F}\cdot\vec{ds} = \int_0^x \mu mg\cos\theta\frac{dx}{\cos\theta}$

$= \mu mg\, x = 20$ J



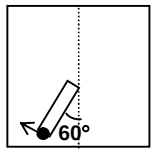
18. $\ell = 2\sqrt{2}h = \frac{1}{2}(g\sin 30^\circ \cos 45^\circ)t^2$

t = 4 sec.



19. Conservation of energy $\frac{1}{2}mv^2 - mg\frac{\ell}{2} = mg\frac{\ell}{2} + m\sqrt{3}\left(2\ell\frac{\sqrt{3}}{2}\right)$

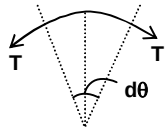
$v = \sqrt{8g\ell}$



20. $s = \left[\int_0^3 (3-t)dt + \int_3^5 (t-3)dt\right] = 6.5\,m$

22. $2T\sin\frac{d\theta}{2} = dmR\omega^2$

$T = \frac{m}{2\pi}R\omega^2$



23. OA $\rightarrow$ acceleration is positive
AB $\rightarrow$ acceleration zero
BCD $\rightarrow$ velocity is decreasing i.e. acceleration is negative

24. for cylinder $V_c = V_0 - \mu gt$ (1)

$\omega = 0 + \frac{2\mu g}{R}t$ (2)

for plank $V_P = \frac{V_0}{2} + \mu gt$ (3)

$V_P = V_C - R\omega$ (4)

$\Rightarrow \quad V_C = \frac{7}{8}V_0$

25. Apply work energy theorem to calculate work done by friction

26. conservation of momentum

$M\sqrt{2gh} - m\sqrt{2gh} = MV_1 + mV_2$ ...(1)

$-1 = \frac{V_1 - V_2}{2\sqrt{2gh}}$ ...(2)

$V_2 = 3\sqrt{2gh}$

$h' = \frac{V_2^2}{2g} = 9h$

28. The particle will leave contact at the same point if $mg - N \le \frac{mv^2}{R}$

i.e N = 0; for $v \ge \sqrt{gR}$

30. apply work-energy theorem

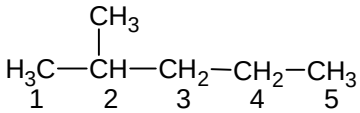
$$mgz - \left[\frac{1}{2}K\left(\frac{mg}{K} + z\right)^2 - \frac{1}{2}K\left(\frac{mg^2}{K}\right)\right] + Fz = 0$$

$\Rightarrow z = 2F/K.$

Chemistry

PART – II

SECTION – A

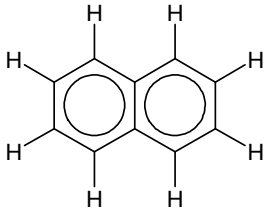
1. 

Iso hexane has 5 type of H for substitution in which C–1, C–3 & C–4 will form optically active compound on mono chlorination.

2. $N_2 + 3H_2 \rightleftharpoons 2NH_3$

$$K_c = \frac{\left(\eta_{NH_3}/V\right)^2}{\left(\eta_{N_2}/V\right)\left(\eta_{H_2}/V\right)^3} = \frac{\left(\eta_{NH_3}\right)^2}{\left(\eta_{N_2}\right)\left(\eta_{H_2}\right)^3}V^2$$

3. $$\left[N_2O_5\right] = \frac{\text{Rate}}{K} = \frac{2.4\times10^{-5}}{6\times10^{-5}} = 0.4 \text{ mol } L^{-1}$$

6. 

7. $$x_2 = \frac{n_2}{n_1+n_2}$$

Mass of solution = $n_1M_1 + n_2M_2$

Volume of solution = $\frac{\left(n_1M_1+n_2M_2\right)}{\rho}$

$$M = \frac{n_2}{V} = \frac{n_2}{\left(n_1M_1+n_2M_2\right)\rho} = \frac{x_2}{x_1M_1+x_2M_2}$$

8. $$r \propto \frac{1}{\sqrt{M}}$$

$M_{CO_2} = M_{N_2O}$

9. $MX_2 \rightleftharpoons M^{2+} + 2X^-$

s 2s

$K_{sp} = 4 \times 10^{-6} = 4s^3$

$s = 10^{-2}$ mol/lit

$s = 200 \times 10^{-2}$ g/L = 2g/lit

12. $Mn^{2+} = 3d^5 4s^0$

$Fe^{2+} = 3d^6 4s^0$

$Co^{2+} = 3d^7 4s^0$

$Ni^{2+} = 3d^8 4s^0$

13. $$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$$

Initial mol.	2	1	0
Eq. conc.	$\frac{(2-2x)}{2}$	$\frac{(1-x)}{2}$	$\frac{2x}{2}$

$K_2Cr_2O_7$ oxides, SO_2 remaining of equilibrium, Hence
Equivalent of SO_2 left = equivalents of $K_2Cr_2O_7$ used
$(2 - 2x) \times 2 = (0.2 \times 0.1 \times 6)$
This gives, $x \approx 0.97$

Now, $K = \dfrac{\left(\frac{2x}{2}\right)^2}{\left(\frac{2-2x}{2}\right)^2 \times \left(\frac{1-x}{2}\right)} \approx 6.96 \times 10^4$

Hence (C) is correct.

14. Number of radial node in 3p orbital = 3 – 1 – 1 = 1 is its radial probability curve will meet the x – axis only at one point and this is the case in curve (A) and (C) is either for 3p or 2s orbital.
Also, as the shell number increases, the distance of maximum probability increases from the nucleus. Hence (C) is correct.

15. In, SO_3^{2-}, S has 3 bond pairs and one lone pair and hence, it is pyramidal in shape not tetrahedral.
Hence (B) is correct.

16. (A). Three possible spin quantum number means an orbital can have at the most three electrons.
So 4th period involves,
3s,3d and 4p subshells.
So, maximum of electrons (and hence elements) = $4s^3 3d^{15} 4p^9 = 3 + 15 + 9 = 27$
(B). All elements of VIIIth group (i.e. 8, 9, 10 groups) are called typical transition metals. Pt also belongs to 10th group. So it is a typical transition metal.
(C). E.C. of Gd (64) = $[Xe]4f^7 5d^1 6s^2$
So number of f – electrons = 7
(D). Due to ineffective shielding by 3d – electrons in Ga, size of Ga is smaller (almost equal) to that of Al.
Hence (C) is correct.

17. A mixture of NH_4OH and $(NH_4)_2SO_4$ is a basic buffer.

$$pOH = pk_b + \log\frac{(NH_4^+)}{(NH_4OH)}$$

$p^K{}_b$ (NH_4OH) = 14 – 9.26 = 4.74
pOH = 14 – 8.26 = 5.74
Let, moles of $(NH_4)_2SO_4$ to be added = x

$$5.74 = 4.74 + \log\frac{2x}{5 \times 10^{-3}}$$

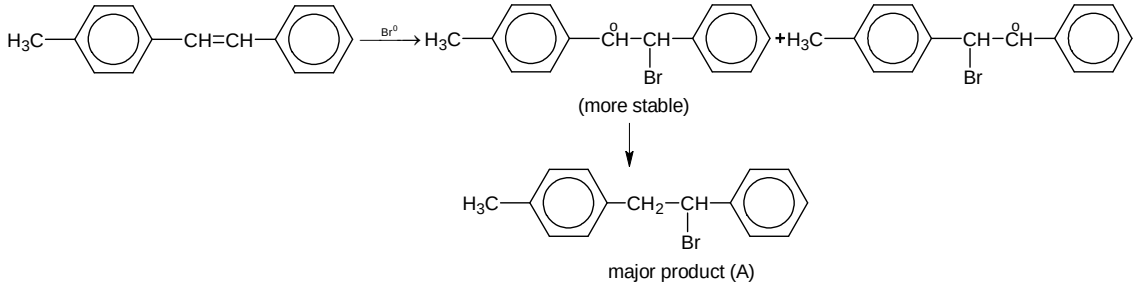
$\therefore$ x = 0.025 mole
Hence (B) is correct.

18. The following equilibrium shows that

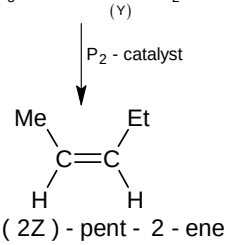
$$NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^-$$

So, NH_4^+ is a stronger acid than H_3O^+.
Hence (B) is correct.

19. It is a case of free radical addition to C = C bond.



20. $Mg_2C_3 \xrightarrow[(X)]{H_2O} CH_3-C\equiv C-H \xrightarrow[(ii)CH_3-CH_2-Cl]{(i)\ Na} H_3C-C\equiv C-CH_2-CH_3$ (Y)



(2Z) - pent - 2 - ene

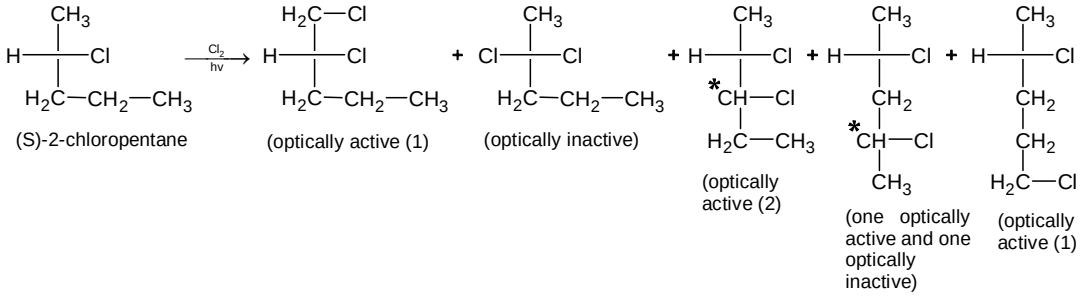
Hence (D) is correct.

21. Since melting point of K is least among all, so heat released in the reaction immediately melts its surface and catches fire.
Hence (C) is correct.

22. $BCl_3 + 3NH_4Cl \xrightarrow{140^0C} \underset{\substack{(A)\\(trichloroborazine)}}{B_3N_3H_3Cl_3} \xrightarrow[(reduction)]{NaBH_4} \underset{\substack{(Borazine\ or\\Inorganic\ benzene)}}{B_3N_3H_6} \xrightarrow{H_2O} NH_3 + \underset{(B)}{H_3BO_3} + H_2$

Hence (C) is correct.

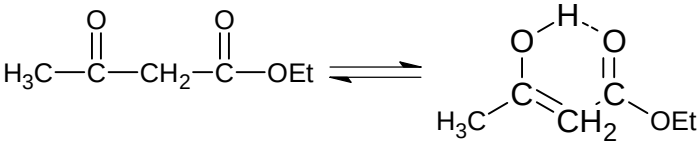
23.



(S)-2-chloropentane
(optically active (1)
(optically inactive)
(optically active (2)
(one optically active and one optically inactive)
(optically active (1)

Hence (A) is correct.

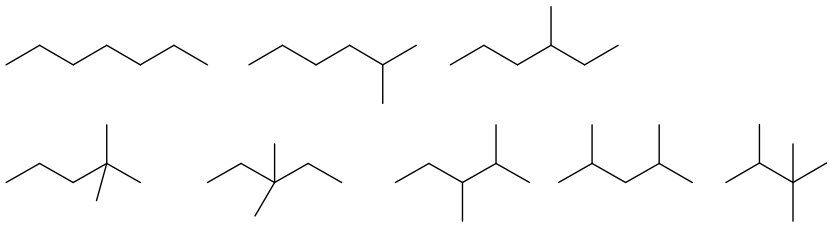
24. Stability due to H – boding is present in ethyl acetoacetate.



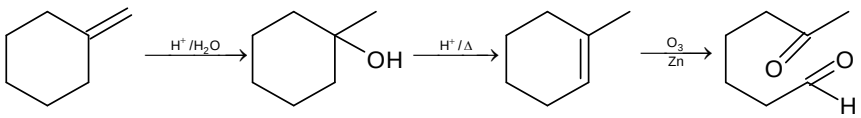
25. The limiting reagent here is HCl. So calculating with the limiting reagent, the total number of produces is 7.5.

26. It may be possible that few particles of gas A have higher velocities than that of few particles of gas B.

27. 96 ppm of SO_4^{2-} = 100 ppm of $CaCO_3$ so 384 ppm of SO_4^{2-} = 400 ppm of $CaCO_3$.

28. 

29. LiF is insoluble in water but not because of its covalent character rather its high lattice energy which is not easily compensated by its hydration energy.

30. 

(Major)

Mathematics PART – III

SECTION – A

1. Here, $P \equiv (6, 6)$
It lies on the circle, thus $72 + 12(g + f) + c = 0$ ..... (1)
Since $y = x$ touches the circle, hence $(g + f)^2 = 2c$ ..... (2)
Solving (1) and (2), $c = 72$

2. Let, $S = 1 + 2\alpha + 3\alpha^2 + ... + n\alpha^{n-1}$
$\Rightarrow \alpha S = \alpha + 2\alpha^2 + 3\alpha^3 + ... + (n-1)\alpha^{n-1} + n\alpha^n$
On subtracting, $S(1-\alpha) = 1 + [\alpha + \alpha^2 + ... + \alpha^{n-1}] - n\alpha^n$
$\Rightarrow S = -\dfrac{n}{1-\alpha}$

3. Put $x = \omega, \omega^2$ ..... (1)
$2^{2010} = a_0 + a_1\omega + a_2\omega^2 + ...$ ..... (2)
$2^{2010} = a_0 + a_1\omega^2 + a_2\omega + ...$
Adding (1) and (2) we get, $2 \times 2^{2010} = 2a_0 - a_1 - a_2 + 2a_3 - a_4 - a_5 + 2a_6 - .....$
$\Rightarrow 2^{2010} = a_0 - \dfrac{1}{2}a_1 - \dfrac{1}{2}a_2 + a_3 - \dfrac{1}{2}a_4 - \dfrac{1}{2}a_5 + a_6 - .....$

4. Total number of required words = 7! – 6! – 6! + 5!

5. $\cos^2 A + \cos^2 B - (1 - \cos^2 C) = 0$
$\cos^2 A + \cos^2 B - \sin^2 C = 0$
$\cos^2 A + \cos(B + C)\cos(B - C) = 0$
$\Rightarrow 2\cos A \cos B \cos C = 0$
Hence, either A or B or C is 90°

6. Here, $a^2 = b^2 + c^2$
$$\sin C = \frac{c}{a} = \frac{2\tan\frac{C}{2}}{1+\tan^2\frac{C}{2}}$$
$$\Rightarrow c\tan^2\frac{C}{2} - 2a\tan\frac{C}{2} + c = 0$$
$$\Rightarrow \tan\frac{C}{2} = \frac{a \pm b}{c}$$
$$\therefore \tan\frac{C}{2} = \frac{a-b}{c} \left(\because \text{for } \frac{a+b}{c},\ C > \frac{\pi}{2}\right)$$

7. $\cos^{-1} x + \cos^{-1} 2x = -\pi$, which is not possible

8. PA + PB is minimum if PA + PB = AB
$$\therefore \begin{vmatrix} 3 & -4 & 1 \\ 1 & 2 & 1 \\ 2k-1 & 2k+1 & 1 \end{vmatrix} = 0$$
$$\Rightarrow k = \frac{7}{8}$$

9. Here, $y = xr$, $z = xr^2$
Also, x, 2y and 3z are in A.P.
Hence, $4y = x + 3z$
$\Rightarrow 4xr = x + 3xr^2$
$\Rightarrow 3r^2 - 4r + 1 = 0$
$\Rightarrow r = \frac{1}{3}$

10. $|z^2 - 3| \geq |z|^2 - 3$
$3|z| \geq |z|^2 - 3$
$|z|^2 - 3|z| - 3 \leq 0$
$0 < |z| \leq \frac{3+\sqrt{21}}{2}$

11. $x^2 - x - a = 0$, $D = 1 + 4a$ = odd
D must be perfect square of some odd integer
Let $D = (2\lambda + 1)^2$
$\Rightarrow 1 + 4a = 1 + 4\lambda^2 + 4\lambda$
$\Rightarrow a = \lambda(\lambda + 1)$
Now, $a \in [6, 100]$
$\Rightarrow a = 6, 12, 20, 30, 42, 56, 72, 90$

12. Equation of normal $y = mx - 2am - am^3$
Put $y = 0$, we get $x_1 = 2a + am_1^2$
$x_2 = 2a + am_2^2$
$x_3 = 2a + am_3^2$
The normal passes through (h, k)
$\Rightarrow am^3 + (2a - h)m + k = 0$
$\Rightarrow m_1 + m_2 + m_3 = 0$
$m_1m_2 + m_2m_3 + m_3m_1 = \frac{2a-h}{a}$
$\Rightarrow m_1^2 + m_2^2 + m_3^2 = -\frac{2(2a-h)}{a}$
$\Rightarrow x_1 + x_2 + x_3 = 6a - 2(2a - h) = 2(h + a)$

13. Any point on the ellipse is $(2\cos\theta, \sqrt{3}\sin\theta)$
The focus on the positive x-axis is (1, 0)
Given that $(2\cos\theta - 1)^2 + 3\sin^2\theta = \frac{25}{16}$
$\Rightarrow \cos\theta = \frac{3}{4}$

14. Given curve is $\frac{(x-2)^2}{4^2} - \frac{(y-2)^2}{4^2} = -1$
So, $a = b = 4$
$\Rightarrow e = \sqrt{2}$

15. Let the vertices be $z_0, z_1, z_2, z_3, z_4, z_5$ w.r.t. centre O at origin and $|z_0| = \sqrt{5}$
Now, ΔOA_2A_3 is equilateral

$\Rightarrow A_2A_3 = \sqrt{5}$

$\Rightarrow$ Perimeter = $6\sqrt{5}$

16. $N = 12600 = 2^3 3^2 5^2 7^1$
Number of even divisors = (3)(2 + 1)(2 + 1)(1 + 1) = 54

17. Here, $y^2 = xz$ ..... (1)
Also, $a^x = b^y = c^z = \lambda$ (Say)
$\Rightarrow x = \frac{\log\lambda}{\log a}$, $y = \frac{\log\lambda}{\log b}$, $z = \frac{\log\lambda}{\log c}$
Putting these values in (1), we get $\log_b a = \log_c b$

18. Given $\frac{ar(r^{10}-1)}{r-1} = 18$ ..... (1)
Also, $\frac{\frac{1}{ar}\left(1-\frac{1}{r^{10}}\right)}{1-\frac{1}{r}} = 6$
$\Rightarrow \frac{1}{a^2r^{11}}\frac{ar(r^{10}-1)}{r-1} = 6$ ..... (2)
From equation (1) and (2), we get $\frac{1}{a^2r^{11}}\cdot 18 = 6$
$\Rightarrow a^2r^{11} = 3$
Now, $P = a^{10}r^{53} = 3^5 = 243$

19. Let α be the root of $x^2 - x + m = 0$ and 2α be the root of $x^2 - 3x + 2m = 0$
Then, $\alpha^2 - \alpha + m = 0$ and $4\alpha^2 - 6\alpha + 2m = 0$
Eliminating α, $m^2 = -2m \Rightarrow m = 0, m = -2$

20. Given equation is, $x^2 - (y + 4)x + y^2 - 4y + 16 = 0$
Since x is real, so, $D \geq 0$
$\Rightarrow (y + 4)^2 - 4(y^2 - 4y + 16) \geq 0$
$\Rightarrow (y - 4)^2 \leq 0$
$\Rightarrow y = 4$
Since equation is symmetric in x and y
Hence, x = 4 only

21. $\frac{1+2\log_3 2}{(1+\log_3 2)^2} + \frac{(\log_3 2)^2}{(1+\log_3 2)^2} = 1$

22. $\frac{(\sin 3\theta + \sin 9\theta) + (\sin 5\theta + \sin 7\theta)}{(\cos 3\theta + \cos 9\theta) + (\cos 5\theta + \cos 7\theta)}$
$= \frac{2\sin 6\theta(\cos 3\theta + \cos\theta)}{2\cos 6\theta(\cos 3\theta + \cos\theta)} = \tan 6\theta$

23. $\left(1-\frac{(s-b)}{s-a}\right)\left(1-\frac{(s-c)}{s-a}\right) = 2$
$\Rightarrow a^2 = b^2 + c^2$
Hence, triangle is right angled

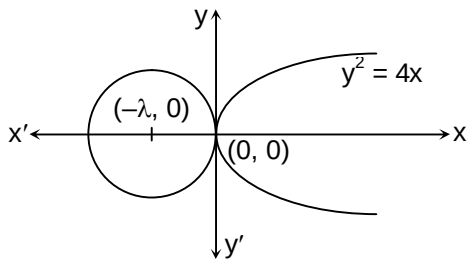
24. $\tan^{-1}\left(\frac{x}{y}\right) - \tan^{-1}\left(\frac{x-y}{x+y}\right)$

$= \tan^{-1}\left(\frac{x}{y}\right) - \tan^{-1}\left(\frac{1-\left(\frac{y}{x}\right)}{1+\left(\frac{y}{x}\right)}\right)$

$= \tan^{-1}\left(\frac{x}{y}\right) - \left(\tan^{-1}1 - \tan^{-1}\frac{y}{x}\right)$

$= \tan^{-1}\frac{x}{y} + \tan^{-1}\frac{y}{x} - \frac{\pi}{4} = \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4}$

25. The graph shows $\lambda > 0$



26. $A \equiv (\alpha, 2\alpha + 3)$, BC = 1 unit. Equation of BC is $y - 3 = 0$
Distance of A from BC is $p = |2\alpha + 3 - 3|$
Area of $\Delta ABC = \Delta = |\alpha|$; $5 \le \Delta < 6$
$\Rightarrow 5 \le |\alpha| < 6$

27. We have, $z^3 + 2z^2 + 2z + 1 = 0$
$\Rightarrow z = -1, \omega, \omega^2$
Since $z = -1$ does not satisfy $z^{1985} + z^{100} + 1 = 0$
While $z = \omega, \omega^2$ satisfy it
Hence, sum $= \omega + \omega^2 = -1$

28. $3^{400} = (81)^{100} = (1 + 80)^{100}$
$= {}^{100}C_0 + {}^{100}C_1 80 + \ldots\ldots + {}^{100}C_{100}80^{100}$
$\Rightarrow$ Last two digits are 01

29. $2 + 3 + 6 + 11 + 18 + \ldots\ldots = (0^2 + 2) + (1^2 + 2) + (2^2 + 2) + (3^2 + 2) + \ldots\ldots$
Hence, $t_{50} = 49^2 + 2$

30. Distribute Rs. 3 to each person
Then, we have to find non–negative integral solution of $y_1 + y_2 + y_3 + y_4 = 6$
Which is given by ${}^{6+4-1}C_{4-1} = {}^9C_3 = 84$